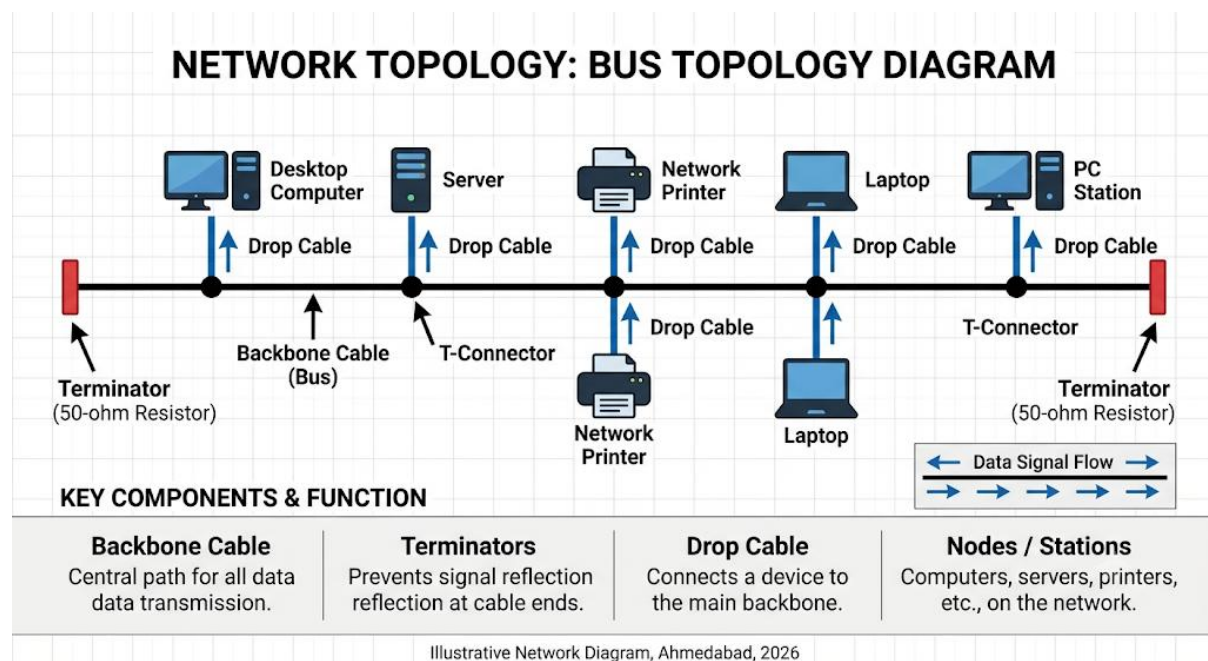


TASK-1



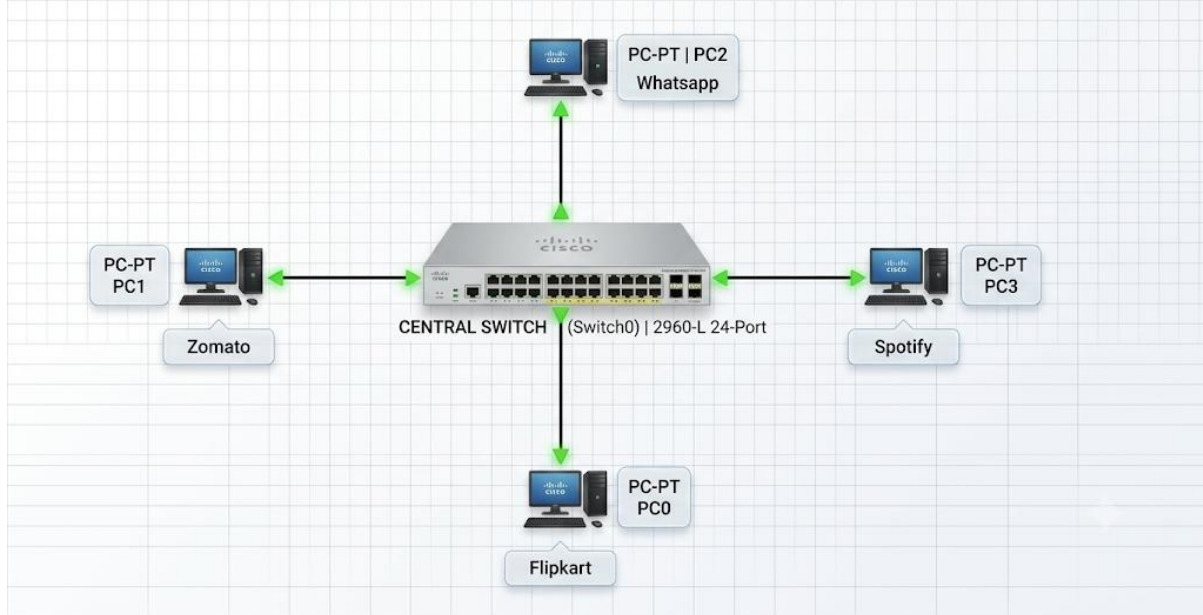
Real-World Example

While obsolete in modern LANs, one classic and significant real-world application of the Bus Topology is the original **Ethernet networks using Coaxial Cables (specifically 10Base2 and 10Base5).**

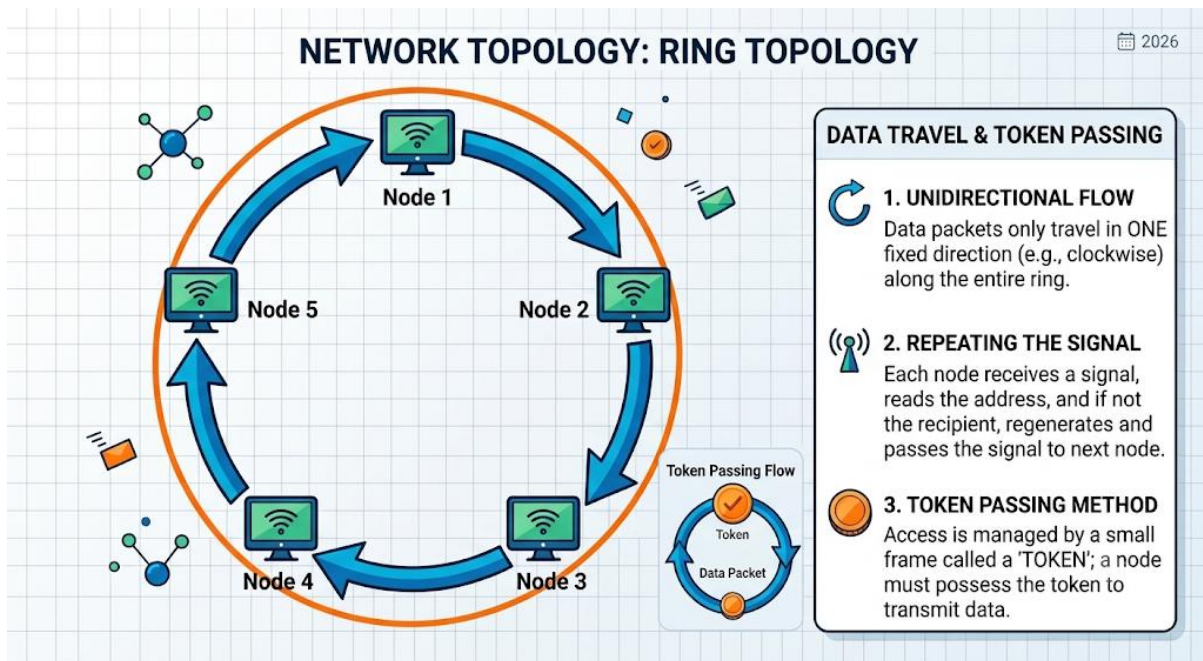
In these early LANs, computers were literally daisy-chained together along a segment of coaxial cable, which acted as the backbone. Terminators were essential components at each end of the coaxial run to prevent signal collision and ensure reliable data transmission across the network segment.

TASK-2

LOGICAL NETWORK TOPOLOGY | STAR CONFIGURATION



TASK-3



TASK-4

Here is a quick comparison between **Mesh** and **Star** topologies outlining two key advantages and two key disadvantages for each:

Comparison: Mesh Topology vs. Star Topology

Topology	Advantages	Disadvantages
Mesh Topology	1. High Fault Tolerance & Redundancy: If one cable or link fails, data simply reroutes through another active link without bringing down the network. 2. High Security & Privacy: Dedicated point-to-point connections between nodes ensure data travels directly to its intended destination without exposure to other devices.	1. High Cost & Cable Density: Requires an extensive amount of cabling and multiple Network Interface Cards (NICs) per device, making installation expensive. 2. Complex Setup & Maintenance: Connecting every device to every other device creates a complicated mesh structure that is difficult to manage and scale.
Star Topology	1. Easy to Expand & Manage: New devices can easily be plugged into the central switch without disrupting the rest of the active network. 2. Centralized Troubleshooting: Because all traffic flows through a central node (switch/hub), isolating and resolving network issues is straightforward.	1. Single Point of Failure: If the central switch or hub crashes, the entire network goes down completely. 2. Dependent on Central Hardware: Overall performance and speed rely heavily on the capacity and quality of the central switch.